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ABOUT ME

Software Engineer with 4 years of experience in the IT field at various companies, including Robert Bosch GmbH, specializing in cloud-native systems and system reliability. Experienced in developing **microservices (Go, JavaScript, Python)** and managing the full application lifecycle using **Docker** and **Kubernetes** across AWS and Azure, implementing **CI/CD pipelines** and enhancing system traceability through observability tools, such as **OpenTelemetry and Prometheus**.

OPENSOURCE CONTRIBUTION

ECLIPSE KUKSA DATABROKER (RUST)

Contributed to the Eclipse kuksa-databroker by integrating **OpenTelemetry for distributed tracing**. This improved system observability, **enabled trace analysis across microservices**, and helped identify latency bottlenecks in automotive data communication.

EXPERIENCE

ROBERT BOSCH MOBILITY — AUG 2025 – MAR 2026

MASTER THESIS: FAULT INJECTION MECHANISMS IN VIRTUALIZED ENVIRONMENTS AND ITS OBSERVABILITY | *Go, Docker, Kubernetes, eBPF, Prometheus, Thanos, AWS (EC2, S3 bucket, Lambda), React, TypeScript, Tailwind CSS, Linux operating system, Windows operating system*

- Deployed cross-platform **observability exporters** (Node, Process, Windows, custom eBPF) as Kubernetes Daemon Sets and utilized kubernetes_sd_configs for dynamic discovery and label-based controls to manage resource overhead.
- Developed a **full-stack observability suite** featuring interactive metrics visualization dashboard with log aggregation and fault Injection.
- Engineered a fault injection framework to introduce controlled resource contention and analyzed its impact on vECU simulations using the observability suite.

ROBERT BOSCH CORPORATE RESEARCH — FEB 2025 – JULY 2025

INTERN | *RUST, Go, Docker, OpenTelemetry, Grafana, WebRTC, Microsoft Azure*

Low-latency communication strategies to enable Vehicle-to-Cloud (V2C) teleoperation. The goal is to reduce end-to-end latency in teleoperation of vehicles by evaluating and integrating WebRTC-based communication for real-time streaming and signaling.

- Evaluating **Microsoft Azure** infrastructure components including:
  - **Proximity placement groups** to ensure physical co-location of compute resources
  - Evaluating the effect of **VNet** Peering for Low-Latency Interconnects
  - Architectural Considerations for Cloud **Scalability** and **Fault Tolerance**
- Analysis of Peer-to-Peer Network Paths Across MNOs, ISPs, and Cloud Infrastructure in Wired and Wireless scenarios.

ROBERT BOSCH CORPORATE RESEARCH — FEB 2024 – NOV 2024

WORK STUDENT | *RUST, Go, Docker, OpenTelemetry, Jaeger, Grafana, Prometheus, eBPF, ROS2, gRPC*

Black box testing of kuksa-databroker using ghz tool. Modified the ghz tool written in **Go language** to capture custom metrics helpful in black box testing to obtain end-to-end latency of **kuksa-databroker**.

White box testing of kuksa-databroker using OpenTelemetry in **rust** programming language.

- Accomplished comprehensive observability of the kuksa-databroker by implementing an **OpenTelemetry** pipeline to track function spans, propagate trace contexts, and optimize data collection, which led to a unified view of application performance for analysis in Jaeger.
- Integrated **Prometheus** for metrics collection to monitor service-level performance and expose custom application metrics alongside tracing data.

This observability setup helps in debugging, monitoring performance, identifying latency bottlenecks, and diagnosing failures by giving a comprehensive view of the application's behavior and interactions.

ENGINECAL TECHNOLOGY PRIVATE LIMITED, BENGALURU **FEB 2022 – FEB 2023**

SOFTWARE ENGINEER | *JavaScript, TypeScript, Python, MEAN stack, Tailwind CSS, AWS, Microsoft Azure, Docker, Kubernetes*

Worked as a Product development Engineer mainly focused on DevOps and Full-stack development.
Developed and maintained a microservices architecture for vehicle data processing with **CI/CD**, deployed on **Azure Kubernetes Service (AKS)** and **AWS Elastic Kubernetes Service (EKS)**.

ELLIPSONIC INDIA SOLUTIONS PVT LTD, BENGALURU **APR 2021 - JAN 2022**

SOFTWARE ENGINEER | *TypeScript, JavaScript, React, Node.js, Prisma ORM, PostgreSQL, AWS, MERN*

Developed and maintained full-stack web applications using the MERN stack (MongoDB, Express.js, React, Node.js).
Managed cloud infrastructure with **AWS EC2**, **AWS Lambda**, **AWS CloudFront**.
Designed a web application to control AWS instances via **AWS CLI** with real-time status updates of EC2 instances. Used **AWS CloudWatch** and **AWS Lambda** for event-driven automation on AWS infrastructure.
Contributed to building a virtual platform for the **Ministry of Health, Singapore**. The platform included the functionality of sending email which made use of **Amazon Simple Email Service (SES), Amazon CloudWatch, AWS Lambda.**
Performed **data engineering** and **backend tasks** in Microsoft Azure, integrating **Prisma ORM** with **PostgreSQL** for database design and queries, and working with **Azure Synapse** Analytics to implement **ETL pipelines** using:

- **Azure Data Factory** for orchestrating data workflows,
- **Azure Data Lake Storage** for scalable storage of structured and unstructured data,
- **Synapse SQL & Spark pools** for data transformation and analysis of large datasets in preparation for analytics workflows.

PROJECTS

Projects	Description
Observability-framework (Master Thesis)	Observability of Fault Injection in Virtualization Environments
FastAPI-OAuth	RESTful API built with FastAPI and SQLAlchemy. This project provides a complete foundation for user management and secure authentication
Repository-context-engine (RAG-pipleine)	A local RAG tool for querying Markdown files, built to eventually serve as a centralized LLM context engine for all your GitHub repositories.
Signaloid-performance-benchmark (GitHub Actions)	Automating performance benchmark of signaloid API using GitHub Actions
Distributed-tracing	Opentelemetry demonstration using custom grpc clients connected to kuksa-databroker

SKILLS

Languages: Python, JavaScript, Go, Rust
Technologies & Tools: FastAPI, Docker, Kubernetes, eBPF, AWS, Microsoft Azure, OpenTelemetry, Prometheus, SQL, NoSQL, MERN Stack, MEAN Stack, gRPC, ROS.

EDUCATION

UNIVERSITY OF STUTTGART **APR 2023 – MAR 2026**

M.SC ELECTRICAL ENGINEERING – SMART SYSTEMS GPA 2.1

Relevant Coursework: Service Management and Cloud Computing, and Evaluation, Advanced Mathematics for Signal and Information Processing, Deep Learning, Detection and Pattern Recognition, Hardware Platforms and Programming of Embedded Systems, Modeling and Analysis of Automation Systems, Industrial Automation Systems.

BANGALORE INSTITUTE OF TECHNOLOGY **AUG 2018 – AUG 2021**

B.E ELECTRONICS AND COMMUNICATIONS GPA 1.5

LANGUAGE PROFICIENCY

English: C2, **German:** A2 (Learning)